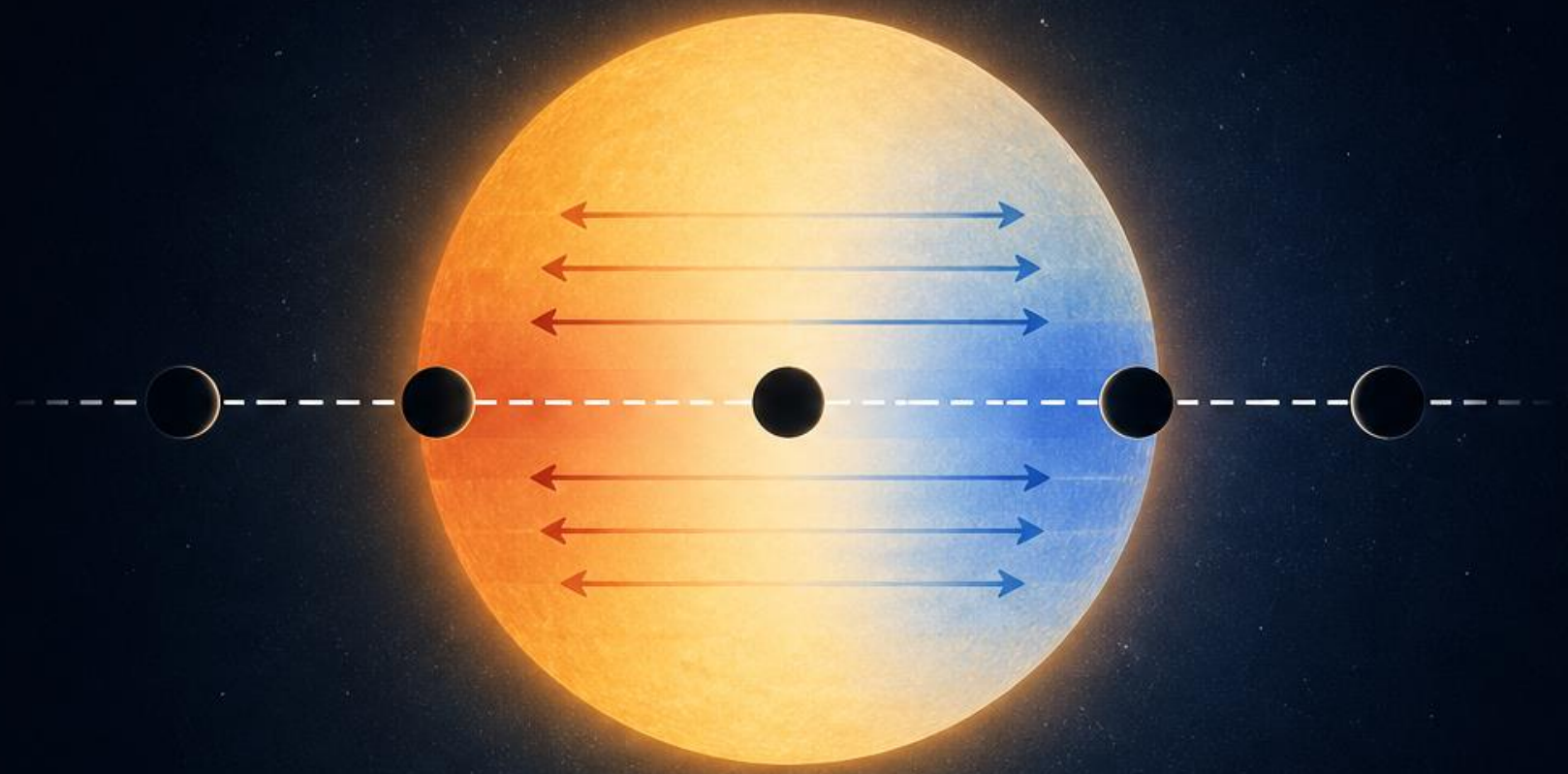
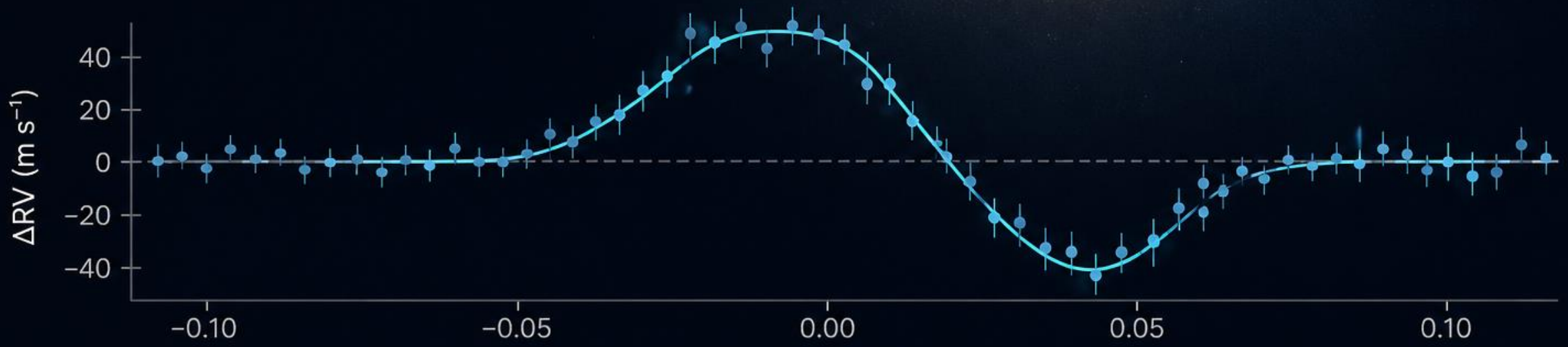
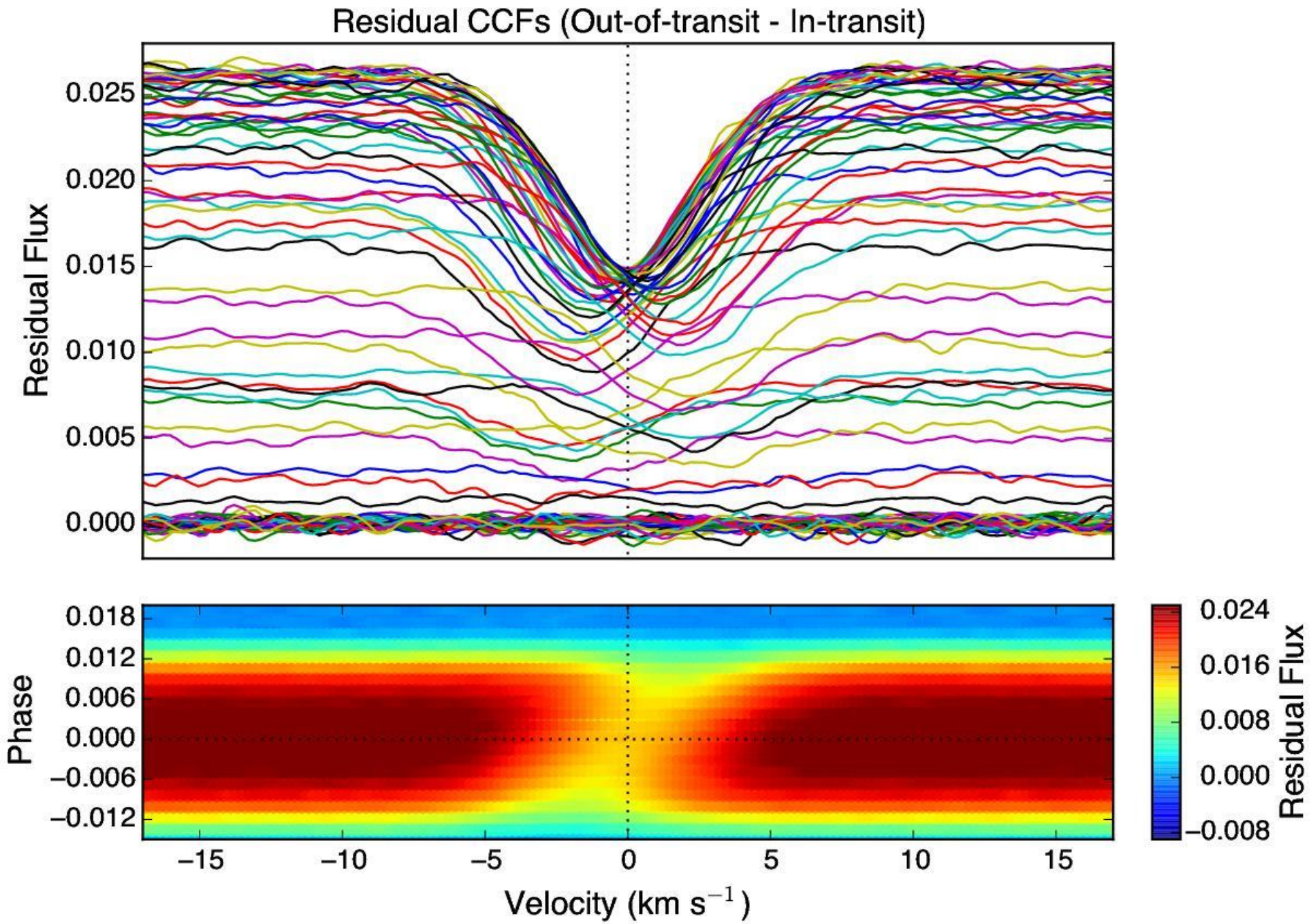
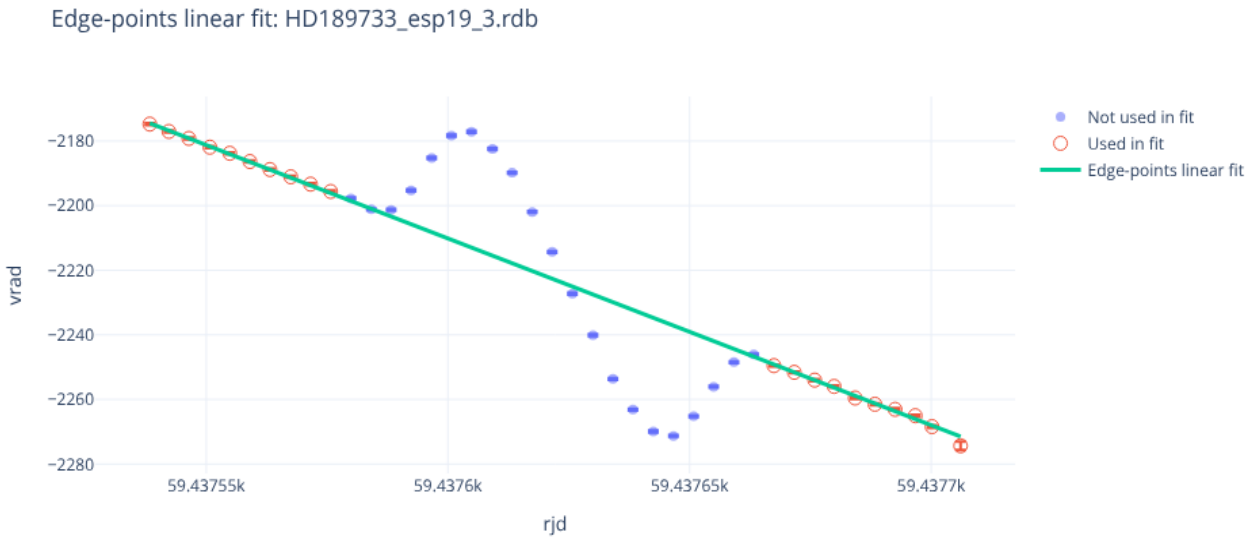
RV from RM

Radial Velocity Estimation using Rossiter–McLaughlin Observations



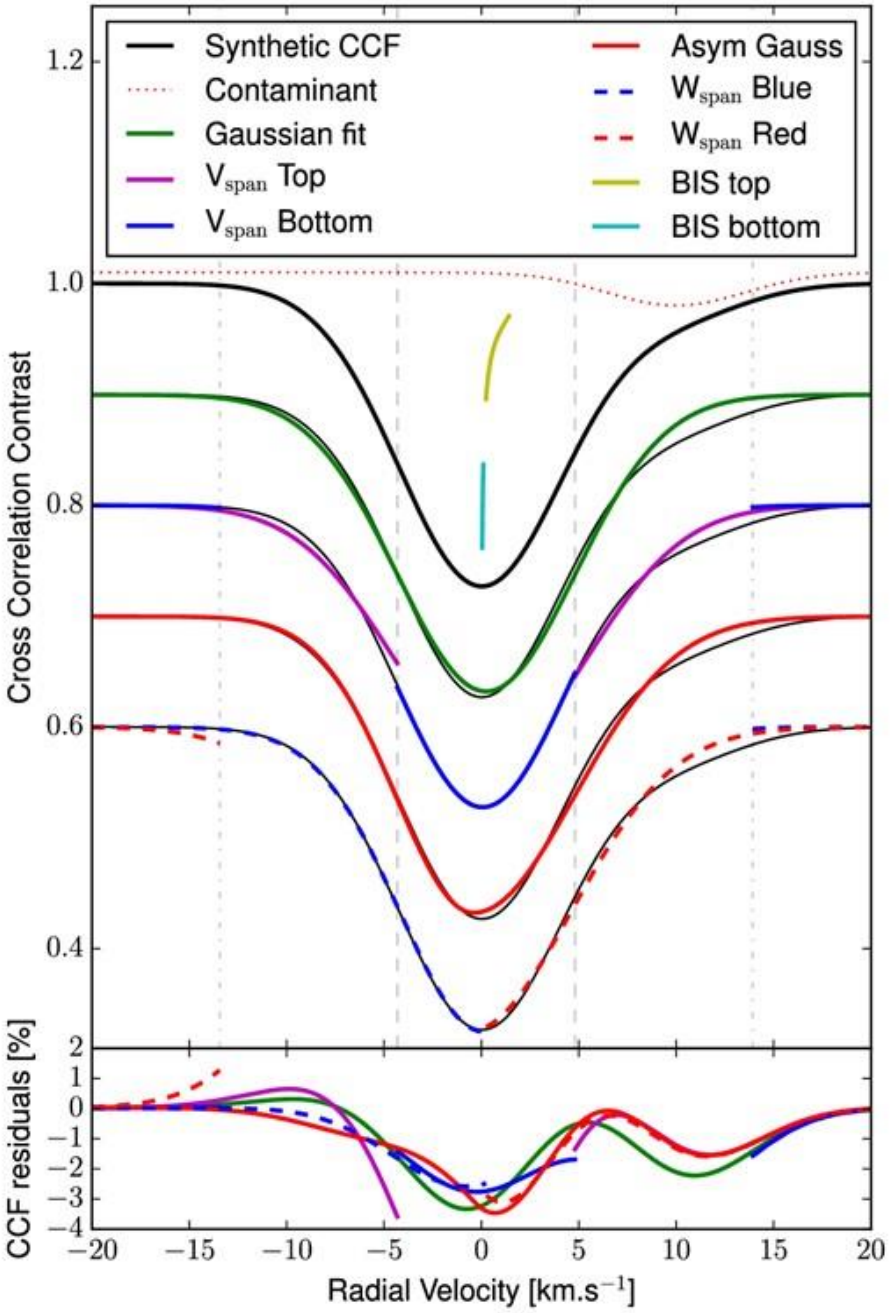


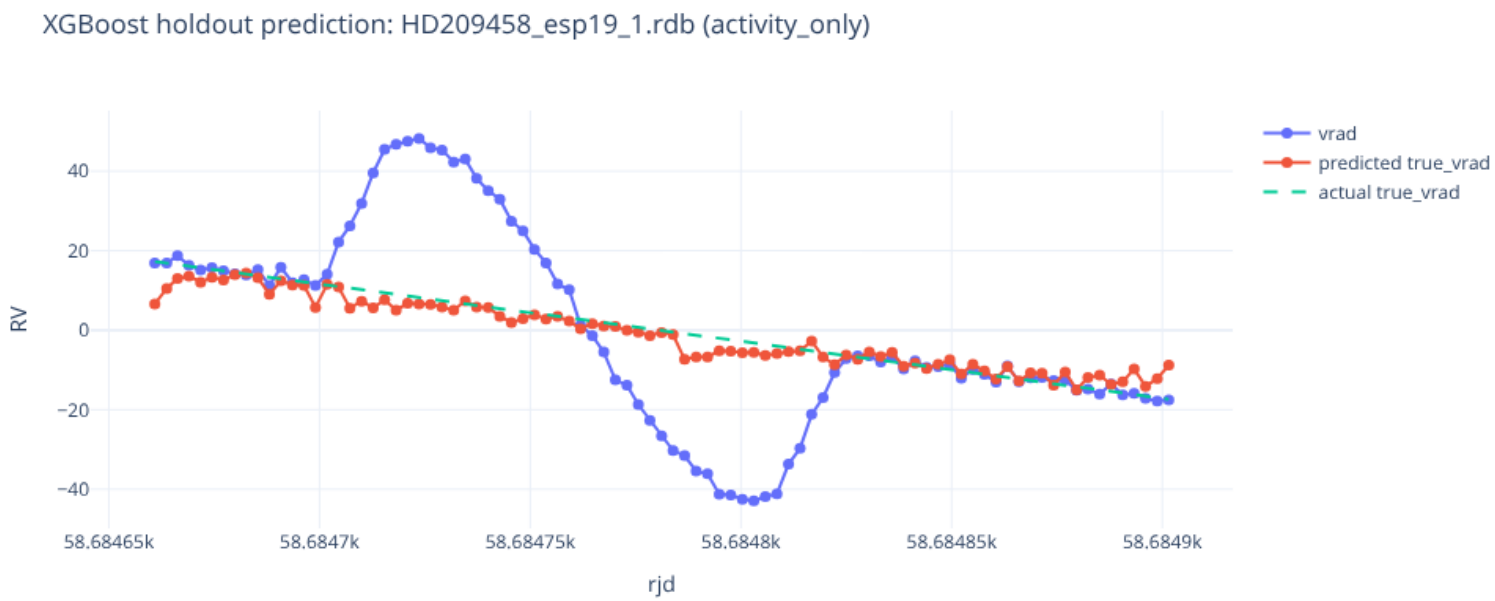

Residual CCFs (Out-of-transit - In-transit)
Residual Flux
0.025
0.020
0.015
0.010
0.005
0.000
Phase
0.018
0.012
0.006
0.000
−0.006
−0.012
−15
−10
−5
0
5
10
15
Velocity (km s^{-1})
0.024
0.016
0.008
0.000
−0.008
Residual Flux

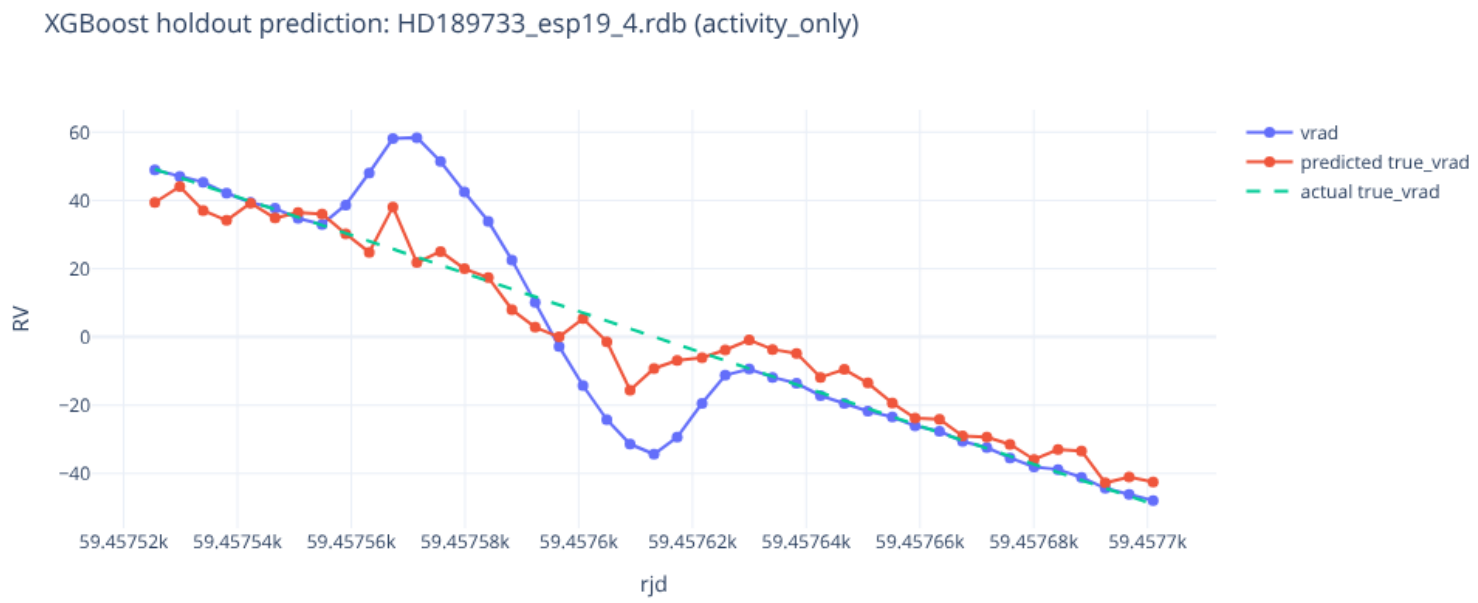


{'HD189733',
'HD209458',
'K2139',
'WASP106',
'WASP118',
'WASP126',
'WASP130',
'WASP166',
'WASP20',
'WASP31',
'WASP50',
'WASP54',
'WASP62'}

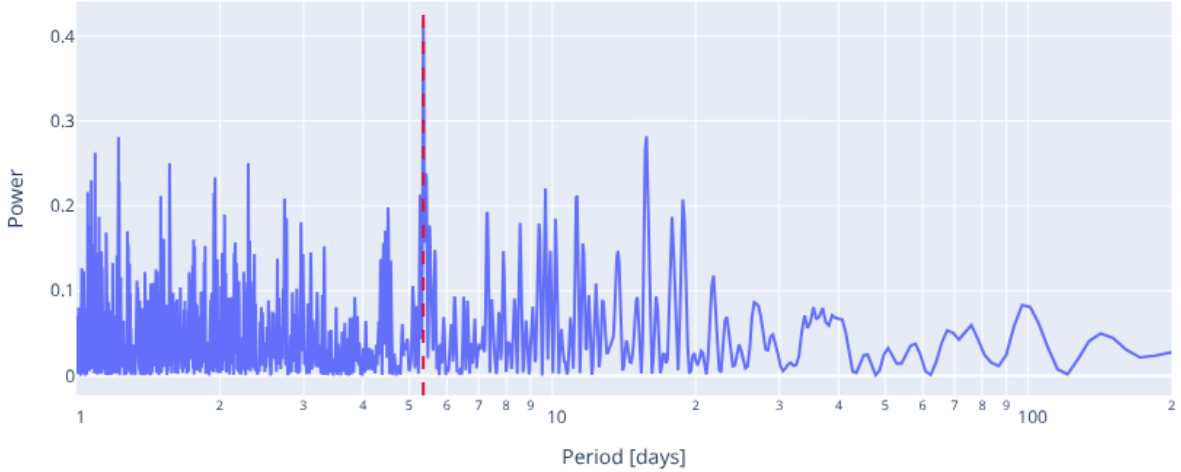
{'HD189733_esp19_3.rdb', 'HD189733_esp19_4.rdb', 'HD189733_harps03_3.rdb', 'HD189733_harps03_4.rdb', 'HD189733_harps03_5.rdb', 'HD209458_esp19_1.rdb', 'HD209458_esp19_4.rdb', 'K2139_esp19_1.rdb', 'WASP106_esp19_1.rdb', 'WASP118_esp19_1.rdb', 'WASP126_esp19_1.rdb', 'WASP130_esp19_1.rdb', 'WASP166_esp19_1.rdb', 'WASP20_esp19_1.rdb', 'WASP31_esp19_1.rdb', 'WASP31_esp19_3.rdb', 'WASP50_esp19_1.rdb', 'WASP54_esp19_1.rdb', 'WASP54_esp19_2.rdb', 'WASP62_esp19_1.rdb', 'WASP62_esp19_2.rdb'}


Synthetic CCF
Contaminant
Gaussian fit
V_{span} Top
V_{span} Bottom
Asym Gauss
W_{span} Blue
W_{span} Red
BIS top
BIS bottom
Cross Correlation Contrast
CCF residuals [%]
Radial Velocity [km.s^{-1}]

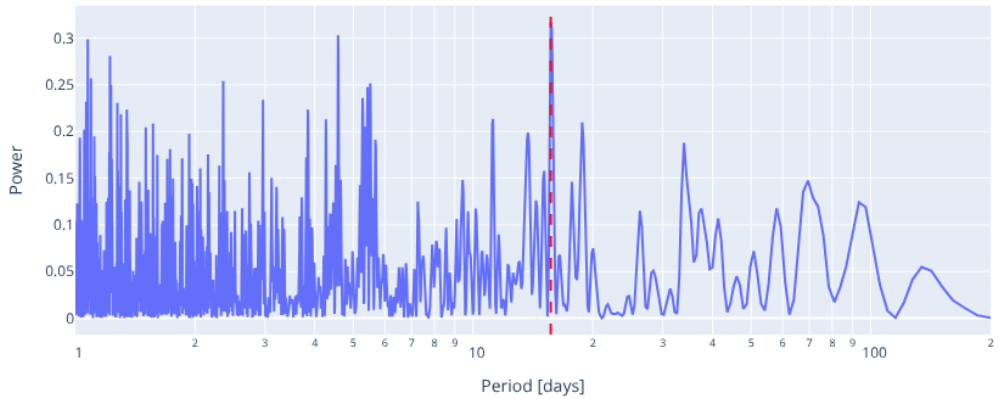

XGBoost holdout prediction: HD209458_esp19_1.rdb (activity_only)
vrad
predicted true_vrad
actual true_vrad
RV
40
20
0
−20
−40
58.68465k
58.6847k
58.68475k
58.6848k
58.68485k
58.6849k
rjd

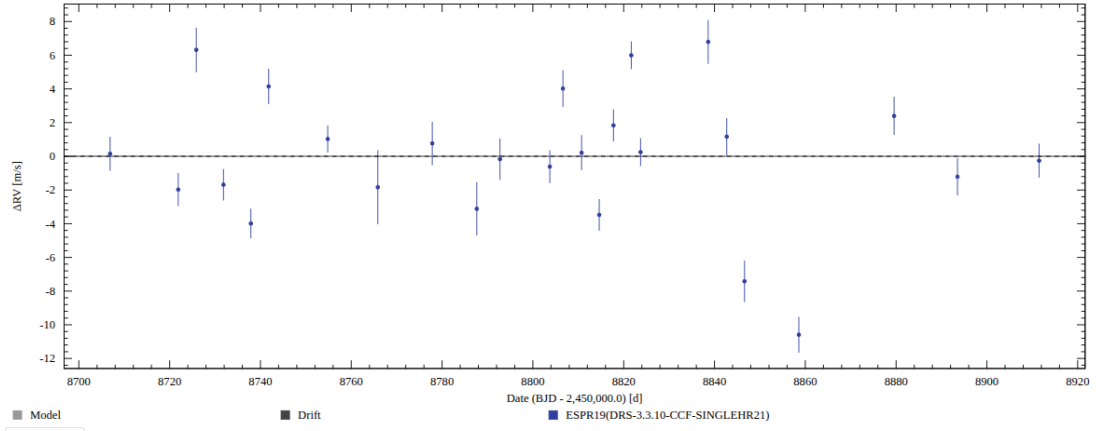

XGBoost holdout prediction: HD189733_esp19_4.rdb (activity_only)
vrad
predicted true_vrad
actual true_vrad
RV
60
40
20
0
−20
−40
59.45752k
59.45754k
59.45756k
59.45758k
59.4576k
59.45762k
59.45764k
59.45766k
59.45768k
59.4577k
rjd

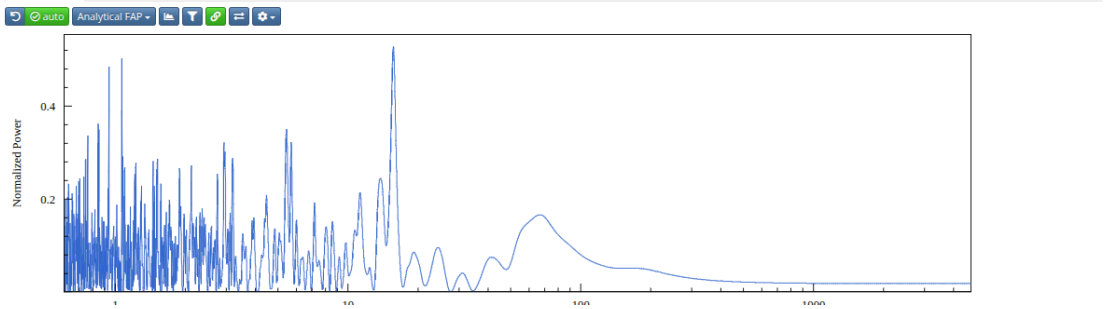
GLS periodogram of predicted RV



GLS periodogram of predicted RV







Name	K [m s^{-1}]	Mass [$M_\oplus$]	Radius [$R_\oplus$]	Period [d]
K2-38 b	$2.95^{+0.44}_{-0.39}$	$7.3^{+1.1}_{-1}$	1.54 ± 0.14	4.01593
K2-38 c	$2.41^{+0.39}_{-0.37}$	8.3 ± 1.3	2.29 ± 0.26	10.56103
TOI-144 b	196.1 ± 0.7	9.89 ± 0.25	–	2088.8
TOI-144 c	1.5 ± 0.2	4.3 ± 0.7	2.11 ± 0.05	6.267852
K2-111 b	2.21 ± 0.32	$5.29^{+0.76}_{-0.77}$	$1.82^{+0.11}_{-0.09}$	5.3518
K2-111 c	$3.27^{+0.31}_{-0.32}$	11.3 ± 1.1	–	15.6785